

Positron Emission Tomography/Computed Tomography

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Accurate anatomical localization of functional abnormalities obtained with the use of positron emission tomography (PET) is known to be problematic. Although tracers such as ^{18}F -fluorodeoxyglucose (^{18}F -FDG) visualize certain normal anatomical structures, the spatial resolution is generally inadequate for accurate anatomic localization of pathology. Combining PET with a high-resolution anatomical imaging modality such as computed tomography (CT) can resolve the localization issue as long as the images from the two modalities are accurately coregistered. However, software-based registration techniques have difficulty accounting for differences in patient positioning and involuntary movement of internal organs, often necessitating labor-intensive nonlinear mapping that may not converge to a satisfactory result. Acquiring both CT and PET images in the same scanner obviates the need for software registration and routinely provides accurately aligned images of anatomy and function in a single scan. A CT scanner positioned in line with a PET scanner and with a common patient couch and operating console has provided a practical solution to anatomical and functional image registration. Axial translation of the couch between the 2 modalities enables both CT and PET data to be acquired during a single imaging session. In addition, the CT images can be used to generate essentially noiseless attenuation correction factors for the PET emission data. By minimizing patient movement between the CT and PET scans and accounting for the axial separation of the two modalities, accurately registered anatomical and functional images can be obtained. Since the introduction of the first PET/CT prototype more than 6 years ago, numerous patients with cancer have been scanned on commercial PET/CT devices worldwide. The commercial designs feature multidetector spiral CT and high-performance PET components. Experience has demonstrated an increased level of accuracy and confidence in the interpretation of the combined study as compared with studies acquired separately, particularly in distinguishing pathology from normal, physiologic tracer uptake and precisely localizing abnormal foci. Combined PET/CT scanners represent an important evolution in technology that has helped to bring molecular imaging to the forefront in cancer diagnosis, staging and therapy monitoring.

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Historically, instrumentation for tomographic imaging of function (single-photon emission computed tomography [SPECT], positron emission tomography [PET]) evolved along a path somewhat different from that of anatomical imaging devices (computed tomography [CT] and magnetic resonance imaging [MRI]) and the corresponding clinical stud-

ies were performed and interpreted separately in different clinical services, ie, nuclear medicine and radiology, respectively. Despite this segregation, the usefulness of combining anatomical and functional planar images was evident to physicians even in the 1960s, preceding the invention of CT. The alignment of tomographic images is a complex procedure owing to the large number of degrees of freedom and, without some common features, such coregistration, may be problematic. In addition to simple visual alignment or the use of stereotactic frames that are undesirable or inconvenient in a diagnostic setting, sophisticated image fusion software was developed from the late 1980s onwards.¹ For (relatively) rigid objects, such as the brain, software can successfully align images from MR, CT, and PET, whereas in more flexible

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environments, such as the rest of the body, accurate alignment is more difficult because of the large number of possible degrees of freedom. Software fusion is also dependent on matching common features that are extracted either from the images or from markers placed on the patient. Functional imaging modalities such as PET and SPECT often lack reliable anatomical correlates and have coarser spatial resolution and greater noise levels than CT or MR.

One way to address the problems of software fusion is by combining devices (emission and transmission) rather than fusing the images post hoc, an approach that has now coined the term hardware fusion. A combined, or multimodality, scanner such as PET/CT can acquire coregistered structure and function in a single study. The data are complementary, allowing CT to accurately localize functional abnormalities and PET to highlight areas of abnormal metabolism. A further advantage of combined instrumentation is that the anatomical images from CT can be used to improve quantitation of functional images through more accurate attenuation, scatter and partial-volume corrections. This is important in achieving accurate and objective assessment of functional parameters such as myocardial perfusion, tumor uptake values and dosimetry for treatment-planning and monitoring response.

Since the commercial introduction of PET/CT in 2001, adoption of the technology has been rapid, particularly in oncology. Advances in CT and PET instrumentation have been incorporated into the very latest PET/CT designs. In this article, we briefly describe some of the early work that led to the commercial exploitation of PET/CT and subsequently to its current designs. The impact of recent advances in CT and PET performance on these designs will be discussed. An algorithm for CT-based attenuation correction (CT-AC) will be described in addition to the challenges that must be addressed by any implementation of the algorithm in practice.

Historical Concepts

The origins of tomographic imaging in medicine date from the 1960s or even earlier, but fusion of tomographic images was not explored systematically until the late 1980s.¹ Following the earlier superposition of planar images, in the 1990s 2 principal approaches have emerged to image fusion: software and hardware. The software approach attempts to align 2 image sets post hoc after they have been acquired on different scanners at different times. In contrast, the hardware approach combines the instrumentation for 2 imaging modalities and thus acquires both image sets within the same reference frame and thereby ensures as accurate alignment as possible.

Image Fusion With Software

Although a complete discussion of the topic is beyond the scope of this chapter, it is instructive to briefly review some of the basic principles of software fusion; a thorough review of software fusion methods can be found in Hawkes et al.² Fusion of 2 image sets is achieved either by identifying common landmarks or fiducials that can then be aligned or by opti-

mizing a metric based on image intensity values. Whatever the method, the number of possible degrees of freedom between the 2 image volumes defines the complexity of the subsequent transformation. For distributions that do not involve a change in shape or size, rigid-body transformations are adequate. When shears (or a nonisotropic dilation without shear) are involved, an affine transformation comprising a linear transformation and translation is indicated. When there are no constraints on the deformation, a nonlinear transformation (warp) is used. Although methods involving the alignment of extracted features or fiducials have shown some success, at least for the brain, most current methods are intensity-based and images are coregistered by assessing the intrinsic information content. Metrics include intensity ratios³ and mutual information.⁴ Although such techniques have shown great success in aligning images of the brain acquired with CT, PET, SPECT, and MR, they have been less successful for other parts of the body. Earlier clinical assessment in the lung⁵ was disappointing, demonstrating a local registration accuracy of 5 to 8 mm, compared with an accuracy of ~2 mm for the brain.⁶ A recent review⁷ suggests that software fusion can achieve an accuracy of 2 to 3 mm for some studies.

Commercially available software has improved considerably during the past several years both in the accuracy of the registration algorithms and in the sophistication of the user interface and display. As an example, Hermes Medical Solutions (Stockholm, Sweden) offers advanced fusion software for many clinical applications, including correction of misalignment errors for PET/CT scans, registration of PET/CT scans with MR scans, registration of longitudinal PET/CT studies, alignment of PET and MR scans in Alzheimer's disease and other forms of dementia, and registration of SPECT or PET myocardial perfusion studies with CT or MR scans of the heart. However, despite considerable progress, fusion software will probably never compete with the simplicity and convenience of coregistered studies acquired on a combined PET/CT scanner.

Multimodality Prototypes

The pioneering work of Hasegawa and colleagues in the late 1980s^{8,9} set the stage for the hardware solution to image fusion. The aim of this work was to design a device that could perform emission (radionuclide) and transmission (x-ray) tomography with the same detector (high-purity germanium operated in fast counting mode).⁹ Although this approach is attractive, the difficulty is to design a detector that does not compromise performance of at least 1 of the 2 modalities. The work was significant, however, in that it highlighted the strengths of a single device that can perform both anatomical (CT) and functional (SPECT) imaging.¹⁰ Of equal significance was the use of the CT images to generate attenuation correction factors for the emission data.¹¹ However, recognizing the difficulty of building a detector that would operate optimally for both CT and SPECT, Hasegawa turned to a different design that comprised a clinical SPECT gamma camera in tandem with a clinical single-slice CT scanner.¹²

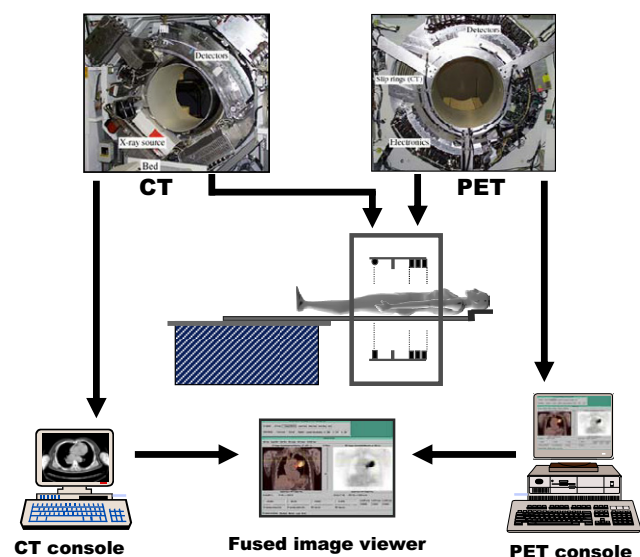


Figure 1 The first PET/CT prototype evaluated clinically at the University of Pittsburgh. The CT and PET components were mounted on a single rotating support and the data acquired from 2 separate consoles. The CT images were transferred to the PET console and then used for CT-based attenuation correction and localization. (Color version of figure is available online.)

The CT scanner (9800 Quick; GE Healthcare, Milwaukee, WI) was positioned in front of, and aligned with, a scintillation camera (600 XR/T; GE Healthcare). The same bed was used to acquire both studies and the images were registered by taking into account the axial displacement between the CT and SPECT imaging fields of view (FOV). After radiotracer injection and usual uptake period, the patient was imaged first in the CT scanner and then in the SPECT scanner. The CT data were used to generate the SPECT attenuation-correction factors. The combined device was used for a limited number of clinical studies, for example, activity measurements for radiation dosimetry in brain tumor patients.¹³

The proposal to combine PET with CT was made in the early 1990s by Townsend, Nutt and coworkers independently of the Hasegawa work. The suggestion was also made to use the CT images to generate the PET attenuation correction factors.¹⁴ The first prototype PET/CT scanner became operational in 1998,¹⁵ designed and built by CTI PET Systems in Knoxville, TN (now Siemens Molecular Imaging) and clinically evaluated at the University of Pittsburgh. The design incorporated a single-slice spiral CT scanner (Somatom AR.SP; Siemens Medical Solutions, Forchheim, Germany) and a rotating ECAT ART scanner (CTI PET Systems, Knoxville, TN). The PET detectors were mounted on the rear of the CT support and the entire assembly rotated as a single unit (Fig. 1). The data processing included an algorithm¹⁶ to scale the CT images from x-ray energy to PET annihilation photon energy (511 keV) and generate the appropriate attenuation correction factors (as discussed below). Over 300 cancer patients were scanned on the prototype and the findings presented in a series of publications.¹⁷⁻¹⁹ The results from the prototype demonstrated the utility of high-resolution ana-

tomic images accurately registered with functional images. The coregistered anatomy localized functional abnormalities and clarified equivocal situations, thus improving the accuracy and confidence of the scan interpretation. The use of a rapidly-acquired, low-noise CT scan in place of a lengthy conventional PET transmission scan improved image quality and reduced scan time.

Current PET/CT Instrumentation

In 1999, GE Healthcare launched a dual-head scintillation camera combined with a low-power x-ray tube and detectors, called the Hawkeye (GE Healthcare).^{20,21} This design features 2 rectangular sodium iodide camera heads with a 350-W x-ray tube. The Hawkeye was the first commercial scanner to offer combined anatomical and functional imaging in a single unit. Then, less than 2 years after the first Hawkeye installation, PET/CT scanners incorporating clinical CT and clinical PET performance became commercially available. The first commercial PET/CT scanner to be announced was the Discovery LS (GE Healthcare) in early 2001. This was followed several months thereafter by the release of the Biograph (Siemens Medical Solutions), and then somewhat later by the introduction of the Gemini (Philips Medical Systems). In the past 6 years, PET/CT designs from all vendors have evolved following advances in CT and PET instrumentation. By 2007, 5 vendors worldwide offered PET/CT designs: GE Healthcare, Hitachi Medical, Philips Medical Systems, Toshiba Medical Corporation, and Siemens Medical Solutions. Current designs offered by Siemens Molecular Imaging, GE Healthcare, and Philips Medical Systems are summarized in Fig. 2. A recent addition to PET/CT designs is the Gemini TF, the first commercial time-of-flight (TOF) PET scanner.²² The Gemini TF has yttrium-doped LSO (LYSO) detectors and is combined with a 16- or 64-slice CT scanner. All designs other than the Discovery LS offer a 70-cm-diameter patient port for both CT and PET. All Gemini and Biograph designs acquire PET data in 3D mode only, whereas the Discovery design includes retractable septa and can acquire data in both 2D and 3D mode.

Advances in Performance for CT and PET

Multidetector CT Scanners

After the appearance of single-slice spiral CT scanners in the early 1990s,²³ CT performance has experienced a resurgence with the advent of multi-detector arrays (MDCT). This was accompanied by increases in x-ray power (60 kW or greater) and computer capacity for data processing and image reconstruction. Dual- and 4-slice CT scanners first appeared about 1998, with scan times of 500 ms, followed by 16-slice (2002) and, more recently, 64-slice (2004) devices. The increasing number of detector rows (slices) has been accompanied by faster rotation times so that state-of-the-art scanners can now achieve a full rotation in as little as 330 ms. Spatial resolution has improved from ~10 lines pairs (lp)/cm in 1990 to 25

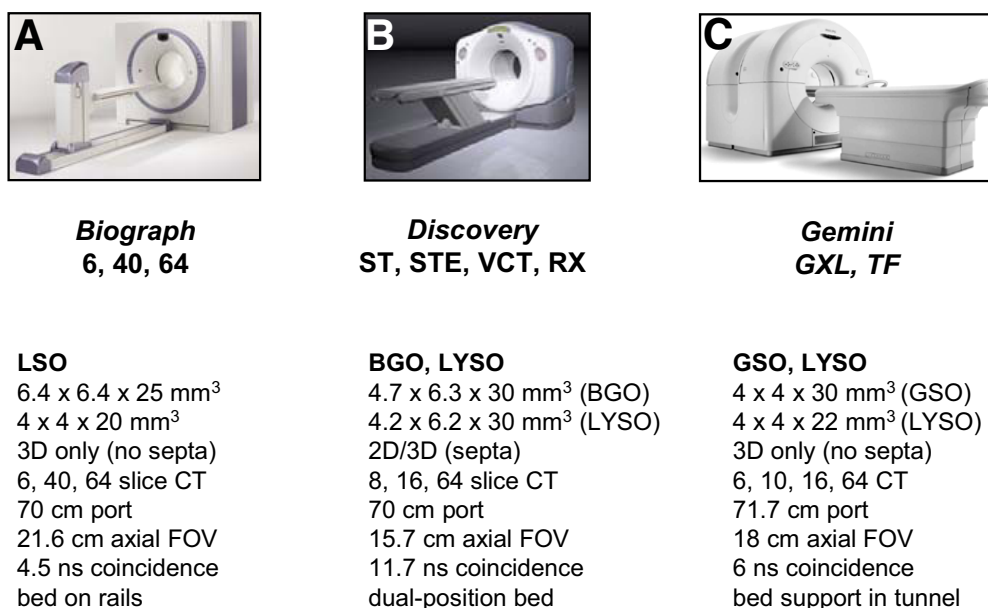


Figure 2 Current PET/CT scanner designs from 3 of the major manufacturers of medical imaging equipment: (A) the Siemens Biograph TruePoint, (B) the GE Healthcare Discovery range, and (C) the Philips Gemini series. (Color version of figure is available online.)

lp/cm or better today, with slice thicknesses less than 1 mm. A significant innovation that will contribute to increased CT performance is the low-weight Straton x-ray tube.²⁴ After many years of slow but steady progress, the past decade has seen significant advances in both hardware and software for CT.

PET Scanners

This section summarizes recent advances in PET technology that define current PET/CT scanner performance.

New Scintillators for PET

For PET detectors, the 1970s saw the transition from thallium-activated sodium iodide (NaI(Tl)) to bismuth germanate (BGO), a scintillator with higher density and photofraction. Although at least one PET scanner design continued to use NaI(Tl) until fairly recently, the majority of PET scanners installed during the 1990s were based on BGO block detectors. In the late 1990s, the introduction of new, faster scintillators such as gadolinium oxyorthosilicate (GSO)²⁵ and lutetium oxyorthosilicate (LSO),²⁶ both doped with cerium, improved the PET-scanner performance. Both GSO and LSO

have shorter decay times than BGO by a factor of 6 to 7, reducing system deadtime and improving count-rate performance, particularly with high activity levels in the FOV. The physical properties of these scintillators are compared in Table 1. Of even more importance for clinical imaging is the potential of faster scintillators to decrease the coincidence timing window, thereby reducing the random coincidence rate. The increased light output of the new scintillators improves the energy resolution because the increased number of light photons reduces the statistical uncertainty in the energy measurement. However, other physical effects contribute to the emission process and the improvement in energy resolution is not a simple function of the number of light photons. The higher light output also increases the positioning accuracy of a block detector, allowing the blocks to be cut into smaller crystals and thereby improving spatial resolution. BGO, LSO, and GSO are not hygroscopic, facilitating the manufacture and packaging of the detectors. GSO is somewhat more fragile and more difficult to machine than either BGO or LSO. LSO has an intrinsic activity concentration (of lutetium-177) of ~280 Bq/mL with single-photon

Table 1 Physical Properties of PET Scintillators

Property	NaI	BGO	LSO	GSO
Density (g/mL)	3.67	7.13	7.4	6.7
Effective atomic number	51	74	66	61
Attenuation length (cm)	2.88	1.05	1.16	1.43
Decay time (ns)	230	300	35 to 45	30 to 60
Photons/MeV	38,000	8200	28,000	10,000
Light yield (% NaI)	100	15	75	25
Hygroscopic	Yes	No	No	No

NaI, sodium iodide; BGO, bismuth germanate; LSO, lutetium oxyorthosilicate; GSO, gadolinium oxyorthosilicate.

emissions in the 88- to 400-keV energy range. Such a radioactive component is of little consequence for coincidence counting at 511 keV, except maybe at very low emission count rates.

Sensitivity

PET is intrinsically a three-dimensional (3D) imaging modality, replacing absorptive physical collimation required for single-photon imaging with the electronic collimation of coincidence detection. The first multi-ring PET scanners incorporated septa, lead or tungsten annular shields mounted between the detector rings. The purpose of the septa was to shield the detectors from photons that scattered out of the transverse plane, restricting the use of electronic collimation to within the plane. Such septa result in inefficient use of the radiation emitted from the patient, but limit scatter and allow 2-dimensional (2D) image reconstruction algorithms to be used. The availability beginning in 1990 of BGO scanners with retractable septa encouraged the use of 3D methodology, at least for the brain, where a 5-fold increase in sensitivity could be realized even with accompanying increases in both scatter fraction and randoms.²⁷ The situation for whole body imaging is far less favorable, in part because of the presence of significant activity just outside the imaging FOV in most bed positions. Instead, particularly for large patients, 2D imaging has been recommended even though higher administered activities of fluorodeoxyglucose (FDG) are required to obtain adequate count rates. This situation changed in the late 1990s with the introduction of LSO- and GSO-based scanners that could be operated with short coincidence time windows (4.5–6 ns) and higher threshold (400–450 keV) energy windows compared with those of a typical BGO scanner, 10 to 12 ns and 350 keV, respectively. Significantly improved whole-body image quality has been achieved in 3D with a 10 mCi (370 MBq) injection of FDG. Since the LSO and GSO scanners have no septa and acquire data in 3D mode only, no comparison can be made with 2D operation. Within the past several years, a limited number of LYSO (lutetium-yttrium oxyorthosilicate)-based scanners with retractable septa have been evaluated in 2D and 3D and recent publications suggest that 3D is now preferred over 2D operation.^{28,29}

The sensitivity of a scanner can also be improved by the addition of more detector material, both in terms of thickness and axial extent. For example, for LSO a 50% increase in thickness (from 20 to 30 mm) results in a 40% increase in intrinsic sensitivity. However, increasing the axial extent by 30% results in a 78% increase in volume sensitivity (for 3D acquisition with no septa). Thus, the latter makes more efficient use of the increased volume of LSO, although there will also be an increase in the number of photomultiplier tubes required (and hence increased cost). Following an injection of a radiotracer such as FDG, the patient potentially receives a radiation dose from all annihilation photons, not just those emitted within the scanner FOV. Therefore, the greater the axial extent (ie, the larger the FOV), the more effective use is made of the emitted radiation and the more efficient use is made of a given volume of scintillator. For most PET/CT

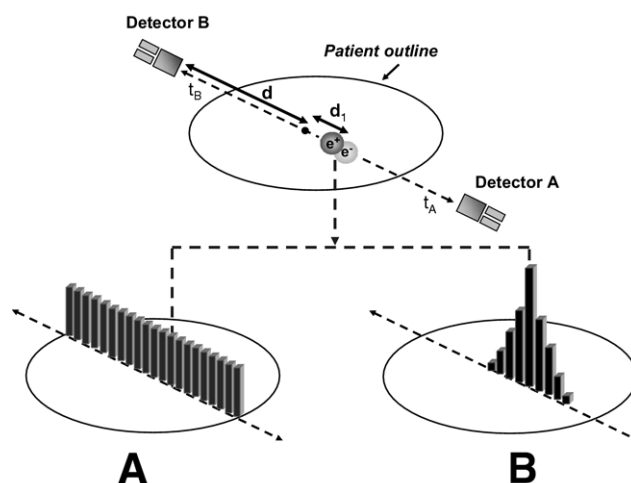


Figure 3 A schematic diagram illustrating PET data acquisition with the incorporation of TOF reconstruction. By measuring the time difference between the arrival of the 2 annihilation photons, the position of the positron annihilation along the LOR can be localized with an accuracy dependent on the precision of the temporal measurement: (A) without TOF information, the annihilation is located with equal probability along the entire LOR; (B) using TOF information, the annihilation point can be localized to a limited range, eg, a 500-ps timing resolution corresponds to 7.5-cm FWHM spatial range.

scanners, axial PET coverage is ~ 16 cm, with one design having an axial extent of 18 cm.²² The most recent design to be announced has an extended FOV covering 21.8 cm axially. The latter comprises more than 32,000, $4 \times 4 \times 20$ mm LSO crystal elements and images 109 2-mm thick transaxial planes per bed position. Data acquisition is fully 3D and the scanner has a peak noise-equivalent count rate (NECR) of ~ 160 kcps.^{30,31}

Signal-to-Noise Ratio

The availability of fast scintillators with high stopping power such as LSO and LYSO has revived interest in time-of-flight (TOF) PET,³² interest that has been further stimulated by the recent announcement of the first commercial PET/CT with TOF, the Philips Gemini TrueFlight (TF).²² The principle of TOF PET is illustrated schematically in Figure 3. Positron annihilation occurs in the patient at a distance $d+d_1$ from one detector and $d-d_1$ from the other detector. The annihilation photons travel at the speed of light (c), the difference in the arrival times at the detectors between the two photons is therefore $2 d_1/c$. Photons originating from the center of the FOV ($d_1 = 0$) obviously arrive in the detectors at the same time. Scanners with fast scintillators and electronics can measure this time difference to within a certain resolution. For example, for a scanner with a coincidence timing resolution of 500 ps, the spatial uncertainty on the position of the annihilation is 7.5 cm. This uncertainty is not sufficient to place the annihilation within a 2-mm voxel (and thereby eliminate reconstruction) but it is superior to having no timing information at all and assigning equal probability of the point of emission to all voxels along the line of response (LOR; Fig.

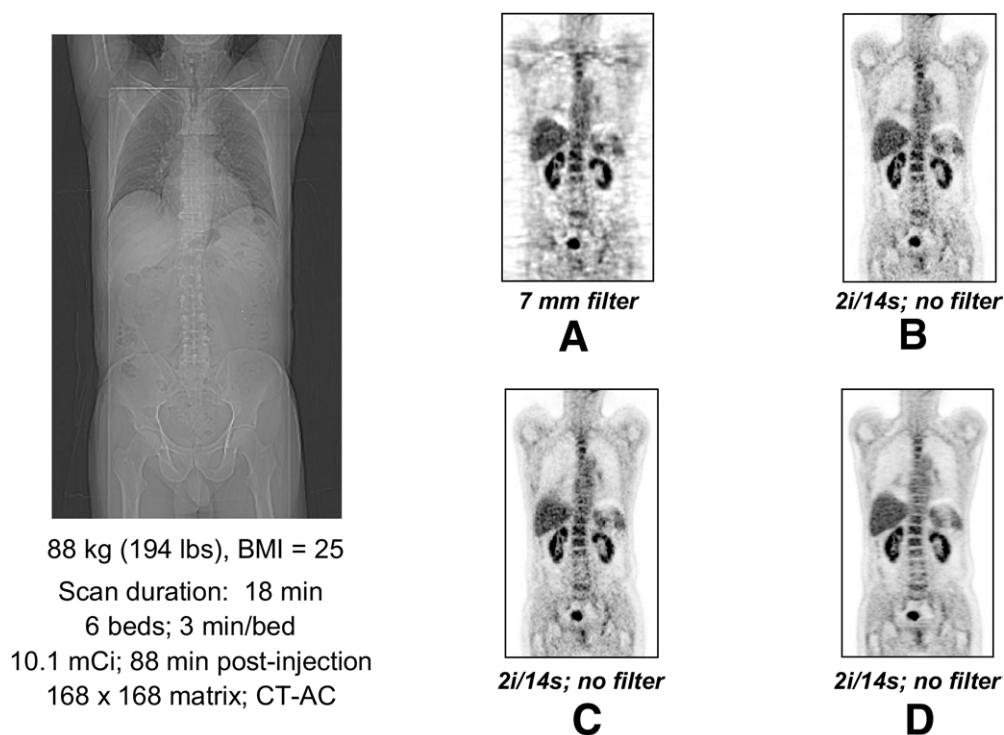


Figure 4 A coronal section of an FDG-PET whole-body scan of a patient with a body mass index of 25 acquired in 3D mode with septa retracted and reconstructed using: (A) 3D filtered back-projection algorithm with reprojection (7-mm Gaussian smooth); (B) FORE + 2D OSEM (14 subsets, 2 iterations; no smoothing); (C) 3D OP-OSEM (14 subsets, 2 iterations; no smoothing); and (D) HD PET: 3D OSEM with PSF reconstruction (14 subsets, 2 iterations, no smoothing). As noted, all reconstructions except 3DRP are unsmoothed.

3A). Instead, the most probable location of the annihilation is at the center of the uncertainty distribution (Fig. 3B). The TOF information is incorporated directly into the reconstruction algorithm, leading to an improvement in the signal-to-noise ratio (SNR). The increase in SNR is proportional to $\sqrt{(D/\delta d)}$, where D is the diameter of the activity distribution and δd is the spatial uncertainty. For a 40-cm diameter uniform distribution of activity and a 7.5-cm uncertainty, the increase in SNR is a factor of ~ 2.3 . As the TOF temporal resolution improves, the spatial uncertainty decreases and the SNR increases by a larger factor. TOF PET was first exploited in the early 1980s³² with scintillators that were fast but did not have good stopping power for 511-keV photons. Interest declined until the recent emergence of scintillators that are both fast and sensitive. The new TOF PET scanners, based on LSO or LYSO, must demonstrate good timing resolution that is stable over time so as to avoid the need for frequent detector re-calibration. Although promising, the clinical impact of TOF PET has yet to be established. A more detailed review of the published contributions to TOF development can be found in.³³

Reconstruction Algorithms

There has been significant progress over recent years in image reconstruction methods through the introduction clinically of statistically based algorithms. Previously, one of the earliest and most widely used 3D reconstruction methods was the re-projection algorithm (3DRP) based on a 3D extension of

standard 2D filtered backprojection.³⁴ Although this algorithm works well for the lower-noise environment of the brain, the quality for whole-body imaging is less than optimal, particularly when rod-source attenuation-correction factors are applied to low-count emission data. Figure 4A, for example, shows a coronal image of a patient with a body mass index of 25 reconstructed using 3DRP. Because CT-based attenuation correction factors have been applied, the image quality is possibly better than would have been obtained with rod-source attenuation-correction factors. The development of Fourier rebinning (FORE)³⁵ was a breakthrough that enabled 3D data sets to be accurately rebinned into 2D data sets and then reconstructed in 2D with a statistically based expectation-maximization (EM) algorithm. However, it was not until the accelerated convergence achieved by the Ordered-Subset EM (OSEM) algorithm³⁶ that iterative methods became of clinical interest. Although FORE and OSEM offer improved image quality compared with 3DRP, the incorporation of attenuation-based weights (AWOSEM), as suggested in the original paper by Hudson and Larkin, further improves image quality. This is demonstrated in Figure 4B, where the same data set as in Figure 4A has been reconstructed with FORE and AWOSEM.³⁷ Further improvement has been achieved by eliminating the rebinning step and implementing OSEM fully in 3D with corrections for randoms, scatter and attenuation incorporated into the system model.^{38,39} The result, again for the same data set, is shown in

Figure 4C. Finally, in a recent development termed, “High-Definition (HD)” PET, the detector spatial response function has also been included in the reconstruction model.⁴⁰ The point spread function (PSF) varies throughout the FOV owing to the oblique penetration of the detectors by annihilation photons (ie, the depth-of-interaction effect). By measuring this variability and then modeling the PSF, improved and near-uniform spatial resolution can be achieved throughout the FOV; the improvement can be seen by comparing Figure 4C with the PSF reconstruction in Figure 4D.

The images in Figure 4 are reconstructed with clinical software provided by a specific vendor (Siemens Molecular Imaging). Of course, most major vendors provide comparable software capable of producing clinical images of high quality. The VUE Point algorithm (GE Health care) is an implementation of 3D OSEM that includes corrections for randoms, scatter, and attenuation and also z-axis smoothing. The Gemini TF (Philips Medical Systems) has TOF capability and therefore the TOF information must be incorporated into the reconstruction.²² For their Gemini scanner, Philips have implemented a distributed list-mode TOF algorithm that is based on a TOF list-mode Maximum Likelihood approach developed by Popescu and coworkers.⁴¹ They previously used a Row-Action Maximum-Likelihood algorithm (RAMLA).⁴² The scatter correction algorithm requires modification to incorporate TOF information. The greatest unresolved effect on image quality, and a challenge to reconstruction algorithms, is now patient size, a significant problem given the current levels of obesity among the US population.

CT-Based Attenuation-Correction Factors

A recognized strength of PET/CT is the availability of CT images for attenuation correction of the PET data,^{16,43} eliminating the need for an additional, lengthy transmission scan. The use of the CT to generate attenuation-correction factors (ACFs) reduces statistical noise. Because the linear attenuation coefficients (μ) are energy-dependent, the CT scan at a mean x-ray energy of ~ 70 keV must be scaled to the annihilation γ -ray energy (511 keV). The mean energy of a polychromatic x-ray beam is defined as the energy of a monochromatic beam that would give the same μ as the polychromatic beam integrated over energy.⁴⁴ The polychromatic beam also undergoes beam hardening, the preferential interaction of lower-energy photons as the beam traverses the body causing the mean energy to increase and the corresponding μ values to decrease.

Energy scaling algorithm for CT. The attenuation of x-rays by tissue depends on the density and the effective atomic number (Z_{eff}) of the absorber material. At CT x-ray energies, the physical processes by which photons are attenuated are the photoelectric effect and Compton scattering. The photoelectric probability varies approximately as Z_{eff}^4 and scales as $1/E^3$, where E is the photon energy. The Compton scattering probability has little dependence on Z_{eff} and decreases lin-

early with $1/E$. The μ for a given material is expressed by the sum of the two components:

$$\mu(E) = \rho_e \{ \sigma_c(E) + \sigma_{\text{ph}}(E, Z_{\text{eff}}) \}$$

where ρ_e is the electron density and σ_{ph} and σ_c are the photoelectric and Compton cross sections per electron, respectively. However, at photon energies above about 100 keV in soft tissue, the photoelectric contribution is essentially negligible compared with the Compton-scatter contribution and therefore the expressions for the μ -values at x-ray energy E_x and γ -ray energy E_γ are:

$$\mu(E_x) = \rho_e \{ \sigma_c(E_x) + \sigma_{\text{ph}}(E_x, Z_{\text{eff}}) \}$$

$$\mu(E_\gamma) = \rho_e \sigma_c(E_\gamma)$$

As a consequence of the 2 separate contributions to $\mu(E_x)$, a single measurement of $\mu(E_x)$ will not uniquely determine $\mu(E_\gamma)$ because, for example, an increase in Z_{eff} could offset a decrease in ρ_e and result in no change in $\mu(E_x)$. In general, therefore, a simple energy scaling of $\mu(E_x)$ is insufficient to yield $\mu(E_\gamma)$. By restricting the problem to biological tissues for which Z_{eff} are all fairly comparable and noting that the contribution from σ_{ph} is relatively small even at x-ray energies, changes in $\mu(E_x)$ are primarily caused by changes in tissue electron density. Thus, for the limited range of biological tissues, a single scaling factor can be used to convert $\mu(E_x)$ to $\mu(E_\gamma)$ for lung, liver, fat, muscle, and other soft tissues. For spongiosa and cortical bone, however, the same scale factor will not apply because of the significant calcium and phosphorous content of bone that result in a higher Z_{eff} different from those of soft tissues.

This issue has been addressed¹⁶ by segmenting bone from soft tissue at a specific threshold and applying different scale factors to the two different tissue classifications—bone and nonbone—corresponding to different values of Z_{eff} . Kinahan and coworkers adopted a threshold of 300 Hounsfield units (HU).¹⁶ Subsequently, Watson and coworkers⁴⁴ proposed a mixture model in which all tissues with μ less than $\mu(\text{water})$ are treated as a mixture of air and water at various concentrations, while tissues with μ greater than $\mu(\text{water})$ are treated as a mixture of water and cortical bone. Since this approach limits the composition to a single value for a given $\mu(E_x)$, a bi-linear scaling function can be defined for biological tissues, as shown in Figure 5. Recent publications on CT-based attenuation correction for PET also propose a “break-point” at 0 HU (corresponding to $\mu(\text{water})$),⁴⁵ although a more appropriate choice may be slightly greater than zero, ~ 60 HU, because some soft tissues and blood conform to the air-water mix but with densities greater than water.

The calibration function has been derived from phantom measurements and has also been validated with patient data.⁴⁶ The calibration of the CT scanner ensures that the soft-tissue values ($\mu < 60$ HU) are independent of the kVp setting of the x-ray tube. This independence does not apply to bone-like tissue ($\mu > 60$ HU) and therefore different calibration functions are required for each kVp setting.⁴⁷

The CT scan is acquired before the emission data so the

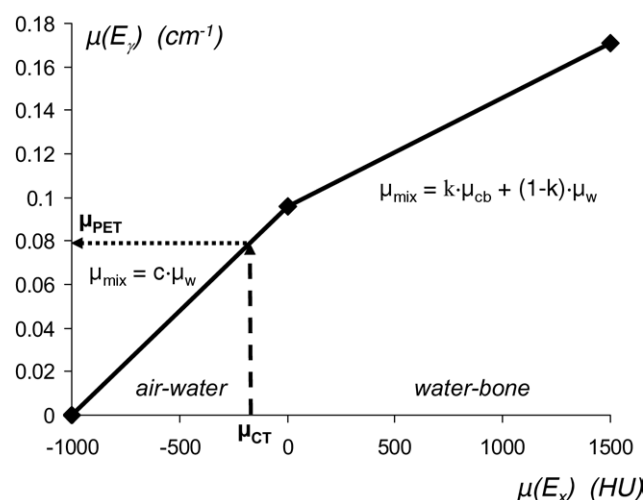


Figure 5 The bi-linear scaling function used to convert CT numbers (Hounsfield Units, HU) to linear attenuation coefficients (μ) at 511 keV. The attenuation-correction factors are generated by reprojecting the μ -map at 511 keV. w = water and cb = cortical bone; k is the concentration of the components of the mixture.

ACFs can be generated for the entire volume. The CT images are first rebinned to the spatial resolution of the emission data. The images are next scaled voxel-by-voxel to the energy of the emission data by applying the bi-linear scaling function (Fig. 5). The scaled CT images are then forward projected to generate ACFs that match the sampling of the emission data. Since the introduction of the PET/CT scanner, CT-based at-

tenuation correction has been a significant focus of research to address the various possible artifacts. The following sections will review the status of this work and the outstanding challenges that remain for CT-based attenuation correction.

Artifacts Specific to CT-Based Attenuation Correction

Although the benefits of CT-based attenuation are now well known and documented, a number of challenges have emerged as the technique has become more widely adopted for PET/CT.^{48,49} There are 2 main concerns (1): the presence of materials in the patient with Z_{eff} values that do not conform to the basic assumptions in the bilinear model and (2) mismatch between the CT and PET caused by patient respiration, cardiac motion, and bowel movement.⁵⁰ Since the first commercial PET/CT installation in 2001, these issues have received considerable attention. Examples of (1) above include metallic objects,^{51,52} dental hardware,⁵³ calcified lymph nodes, and intravenous^{54,55} and oral contrast.^{56,57} Materials with high Z_{eff} may even exceed the dynamic range of attenuation values measurable by CT and severe image artifacts can result. Of particular importance in the assessment of head and neck cancer is the presence of dental fillings.⁵³ A number of metal-artifact reduction techniques have been explored,⁵⁸ including modified reconstruction methods⁵⁹ and segmentation approaches⁶⁰ that can significantly reduce the artifacts.

Some characteristic artifacts associated with CT-based attenuation correction are illustrated in Figure 6. When tidal breathing is adopted for both CT and PET, respiration effects

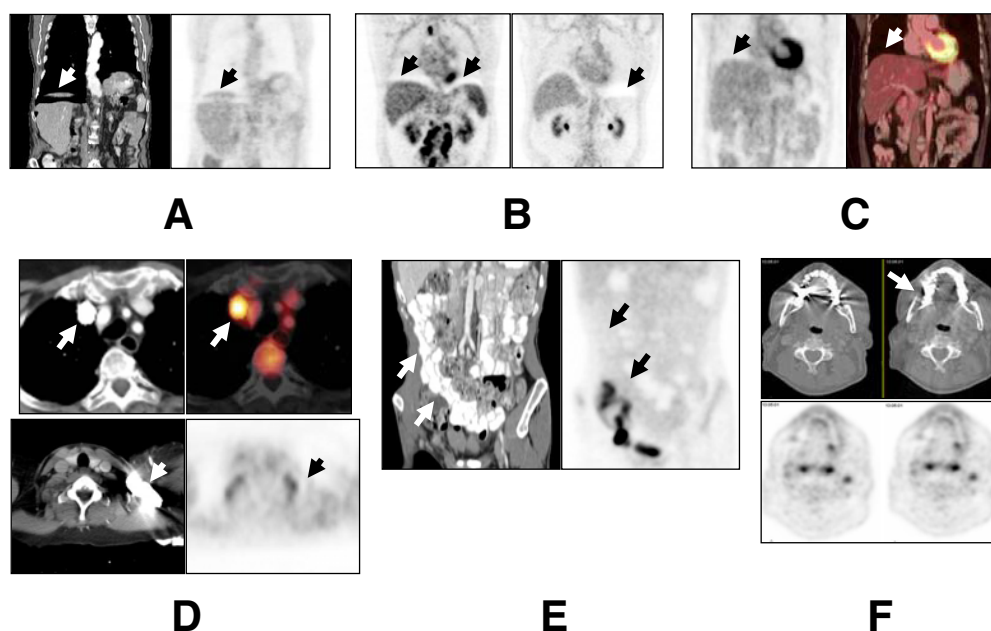


Figure 6 Potential image artifacts generated from CT-based attenuation correction: (A) an artifact caused by respiration in which the dome of the liver is displaced into the base of the right lung; (B) curved photopenic areas above the liver and spleen caused by CT and PET mismatch from respiratory movement of the diaphragm; (C) an example of a well-registered study that is free of artifacts; (D) the variable effects of intravenous contrast showing an artifact on the PET image (top row) caused by a contrast bolus and the absence of an artifact on PET (bottom row); (E) the effect of oral contrast where the presence of contrast in the GI tract does not cause an artifact on the PET image (arrow); and (F) the effect of dental fillings on the CT and PET images.

include an apparent displacement of the dome of the liver into the lower lobe of the right lung (Fig. 6A),⁶¹ creating a corresponding region of apparent activity on the PET scan (arrow). A curved photopenic region at the top of the liver and spleen in the PET image (Fig. 6B) is also observed in some studies. Although such artifacts may occur for any patient following a tidal breathing protocol,⁶² the documented incidence is much lower for faster, higher-performance CT scanners. Figure 6C is an example of a study, acquired on a 6-slice CT scanner, that shows no evidence of breathing artifacts or misregistration. The clinical significance of respiratory artifacts has been studied for an early PET/CT design in a series of 300 patients⁶³ and was found to result in ~2% incorrect diagnoses.

Intravenous Contrast

The use of intravenous contrast may be indicated when the CT scan is performed for clinical purposes as opposed to low-dose CT performed for attenuation correction and localization only. Intravenous contrast contains iodine at concentrations high enough to enhance CT values without a corresponding change in density, and it is used in CT to enhance μ values in the vasculature by increasing the photoelectric absorption compared with the blood. CT contrast results in a 40% change in attenuation at CT x-ray energies whereas at 511 keV where the photoelectric effect is negligible, contrast has only a 2% effect or less.⁶⁴ However, if contrast-enhanced tissue voxels are misidentified as a water-bone mix, the scaling factor will be incorrect and the erroneously scaled pixels may generate artifacts in the PET image (Fig. 6D, top row).⁶⁵ Tens of thousands of PET/CT scans have now been performed in the presence of intravenous contrast and experience has shown that contrast administration does not generally cause a problem that could potentially interfere with the diagnostic value of such scans.^{54,66,67} This is largely due to the fact that intravenous contrast is rapidly dispersed throughout the vascular system. An exception may be the passage of the contrast bolus through a major vessel, although even this does not always generate an artifact on the PET image (Fig. 6D, bottom row). Optimized CT protocols have been developed for the administration of intravenous contrast that avoid most of the foregoing issues.⁶⁸ A recent publication⁶⁹ has documented a rate of 2 patients per 100 where an incorrect management decision would have been made because of the use of noncontrast, low-dose CT acquired for localization and CT-AC only.

Oral Contrast

Oral contrast is administered to enhance radiographic visualization of the gastrointestinal tract. Its distribution is somewhat variable, however, both in spatial distribution and level of enhancement. Modifications to the basic scaling algorithm have been introduced to distinguish oral contrast-enhanced voxels from bone pixels.⁶⁴ As with intravenous contrast, there is little evidence that the presence of oral contrast results in diagnostic errors of any significance.⁷⁰ Figure 6E shows a patient imaged with oral contrast; enhancement of the colon on the CT image (left; arrows) shows no corresponding artifactual uptake on the PET image (right). Nevertheless, in some protocols, contrast CT is performed in

addition to the low dose CT for attenuation correction and localization, thereby increasing the radiation dose to the patient. However, a low-dose whole-body CT in addition to a clinical CT with contrast over a limited axial range (single PET bed position) may involve less radiation dose than a whole-body clinical CT with contrast. Of course, CT AC problems with oral contrast can be eliminated entirely if negative contrast agents such as mannitol or even water are used instead of the usual high-Z agents.⁷¹

Metal Implants

Dental artifacts can be corrected on CT through the use of novel reconstruction techniques,⁵⁹ as shown in Figure 6F. The uncorrected (left) and corrected (right) images for CT (top) and PET (bottom) demonstrate that the reconstruction algorithm significantly improves the CT image without affecting the PET image, verifying that CT-based attenuation correction is actually a robust technique. Although metallic implants such as artificial hip prostheses can sometimes cause artifacts on CT, this appears to be due more to patient movement between the CT and the PET scan than to the presence of prostheses per se.⁷² The nonattenuation-corrected image is, in any case, available to resolve ambiguities.

Respiratory Motion

In recent years, the most widely addressed issue related to CT-based attenuation correction has been respiratory motion⁷³⁻⁷⁶ and the artifacts created by the mismatch between CT and PET.⁷⁷ These issues are addressed in detail in the accompanying article by Nehmeh and Erdi in this volume and are only briefly discussed here. Rotating⁶⁸ Ge sources used in conventional PET scanners results in a transmission scan that averages patient respiration in a way compatible with the corresponding emission scan. The use of CT-AC suggests a number of different protocols must be explored to resolve the mismatch problem. For example, the advent of fast, spiral CT scanners made breath-hold CT a reality, although clinical images are typically acquired with full inspiration to separate lung structures. Such an expansion of the chest does not match a PET scan acquired with shallow breathing and results in serious attenuation correction artifacts in the anterior chest wall. The appearance of artifacts due to respiratory motion and the spatial and temporal mismatch between CT and PET images has led to intensive research to identify the best respiratory protocol. A number of different protocols have been explored, including:

- Continuous shallow breathing for both CT and PET⁷³;
- CT scans acquired over the diaphragm with limited breath hold⁷³;
- Breath-hold CT acquired with partial inspiration⁷³;
- Motion-averaged CT over many respiration cycles^{78,79};
- Cine CT acquiring a full breathing cycle per slice⁸⁰;
- Respiratory-gated CT plus shallow-breathing PET⁸¹;
- Deep-inspiration breath hold^{82,83};
- Breath-hold CT plus gated PET^{84,85};
- Respiratory-gated CT and PET.⁸⁶

Currently, the simplest and most widely used protocol is shallow breathing for both CT and PET.⁷³ Early single- or

dual-slice PET/CT designs exhibited a high incidence of breathing artifacts (Fig. 6A and B).⁶² However, with the incorporation of fast MDCT into PET/CT scanners, the incidence of such artifacts has been greatly reduced. Nonetheless, the CT images still do not exactly match the motion-averaged PET acquisitions and protocols such as slow CT acquisition have also been explored. The clinical significance of these attenuation-correction effects continues to be debated, particularly with respect to lesions in the base of the lung and dome of the liver where curved photopenic areas are observed (Fig. 6B). Displacement of such lesions may result in incorrect localization or, worse, a failure to identify them correctly thus leading to misdiagnosis. Shallow breathing during PET/CT has been shown to be inadequate for the comprehensive staging of lung cancer.⁸⁷ Nevertheless, a significant percentage of studies acquired on even a 6-slice CT scanner show good registration with shallow breathing.

Finally, 2 other effects can also influence the accuracy of CT-AC: the truncation of the transverse FOV⁸⁸ and the presence of scattered radiation. Truncation of the FOV arises because typically CT scanners have a 50-cm diameter and PET scanners a 60- to 70-cm diameter FOV. Simple software extrapolation techniques have proved effective in extending the CT FOV to match that of PET, at least with an accuracy adequate for CT-AC.^{89,90} Scatter is increased by imaging with the patient's arms in the FOV. However, the short scan times achievable with state-of-the-art PET/CT allow almost all patients to tolerate imaging with their arms raised, reducing the effects resulting from increased scatter. (Head-and-neck cancer patients, of course, continue to be scanned with their arms down.)

Despite the artifacts and other issues discussed previously and occasional opinions to the contrary,⁹¹ CT-based attenuation correction has become the de facto standard for PET/CT. The advantages, which include convenience and short acquisition times, largely outweigh the drawbacks. In a small number of studies, quantitative comparisons have been made between ACFs generated from standard PET transmission scans and from CT^{75,92,93} and, even though some differences in SUV values have been noted, nothing of diagnostic significance has been documented.

Radiation Dose Considerations

The radiation exposure to the patient from a PET/CT scan is both external, from the CT scan, and internal, from the injected PET radiotracer,⁹⁴ and has emerged as an issue of some concern.

External Dose

Dose assessment in CT is challenging and depends not only on the body region exposed but also on a variety of scan-specific parameters, including tube potential (kVp), tube current and exposure time (mAs), slice collimation, and pitch.⁹⁵ In addition, the dose also depends on certain technical features of the scanner, such as beam filtration, beam-shaping filter, geometry, and acquisition algorithm. Therefore, values

for CT patient dose vary considerably among centers and among scanners. For whole-body CT scans that extend from the level of the thyroid to the symphysis, the effective CT dose E_{ext} can be estimated approximately as follows:

$$E_{\text{ext}} = \Gamma_{\text{CT}} \cdot \text{CTDI}_{\text{vol}}$$

where $\Gamma_{\text{CT}} = 1.47$ mSv/mGy is the dose coefficient that relates the volume CT dose index CTDI_{vol} to the effective dose. For a typical set of clinical scan parameters, the CTDI_{vol} is 13 mGy⁹⁶ resulting in a total effective whole-body dose of 19 mSv. However, many centers acquire the CT scan for attenuation correction and localization only, reducing the whole-body dose to as low as 3 mSv or even lower. In addition, there are a number of strategies to make more efficient use of the radiation, such as tube current modulation and automatic exposure control.^{97,98}

Internal Dose

The internal radiation dose will depend on the biodistribution and the physical and biological half-life of the radiotracer as well as the activity administered to the patient. The dose is generally expressed as the effective dose and the absorbed doses to the whole body and to the individual major organs. The effective dose E_{int} resulting from administration of activity A of a given radiotracer can be estimated from the following:

$$E_{\text{int}} = \Gamma \cdot A$$

where Γ is a dose coefficient computed for the adult hermaphrodite Medical Internal Radionuclide Dosimetry phantom. Currently, the principal radiotracer of interest is FDG, for which the dose coefficient is 19 $\mu\text{Sv}/\text{MBq}$,⁹⁹ although a higher dose coefficient of 29 $\mu\text{Sv}/\text{MBq}$ has also been published.¹⁰⁰ As for phantom-based dosimetry generally, the foregoing FDG dose coefficient applies to patients whose body habitus reasonably approximates that of the 70-kg adult hermaphrodite phantom. It is neither age- nor gender-specific and does not account for differences among individuals in terms of their FDG pharmacokinetics. Dose coefficients for adult females (based on 54-kg anthropomorphic model) and for younger individuals (based on low body-mass models) are now also available.¹⁰¹ Based on the published value⁹⁹ for the dose coefficient, the average whole-body dose for a typical 10-mCi (370-MBq) administered activity of FDG is 7 mSv. However, most radiotracers do not distribute uniformly in the body, and, for example, the critical (ie, highest-dose) organ for FDG is the bladder due to its primarily urinary excretion.

Total Dose

The total dose for PET/CT is, of course, the sum of the internal and external doses. For a diagnostic CT and FDG PET scan, the effective dose is ~ 25 mSv. However, this can be reduced to 10 mSv or less if a low-dose CT—for localization and attenuation correction only—is acquired. In practice, the PET/CT dose to a specific organ will depend on the exact protocol; for example, if the CT scan does not include the

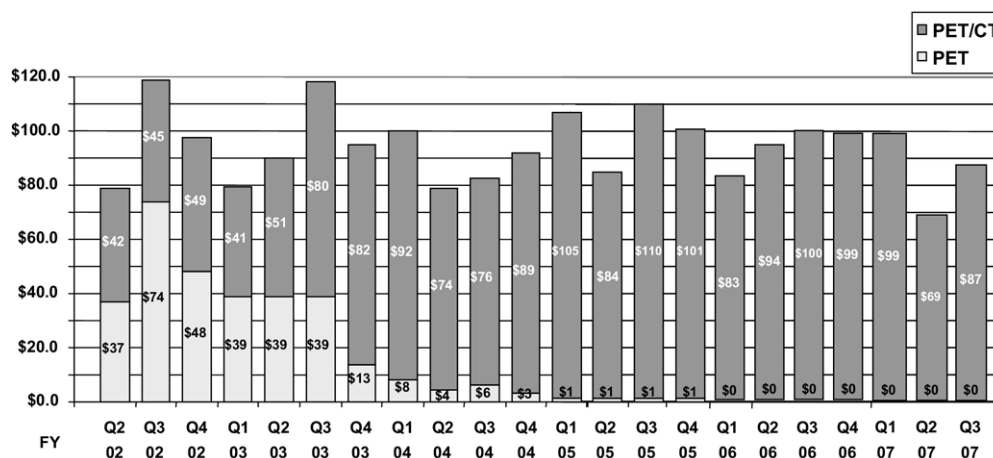


Figure 7 Shipments of PET and PET/CT scanners for the U.S. market as recorded by the Nuclear Equipment Manufacturers Association for the period January 2002 to October 2007. Note that the figures (in \$M) reflect the total revenue for all shipments from which the selling price and individual unit type cannot be determined. Shipments of PET-only scanners declined during this period to zero from January 2006 onwards. The overall market for PET or PET/CT remained fairly constant throughout this period, although since January 2007, with the reduction in reimbursement due to the introduction of the Deficit Reduction Act, sales have declined somewhat.

bladder, the dose to the bladder wall will be due entirely to FDG. For a smaller patient and a higher-sensitivity scanner, a lower FDG dose can be used, potentially limiting the effective dose to 5 mSv or less. In comparison, the world-wide average annual dose due to the natural radioactive background is 2.4 mSv.

The Clinical Role of PET/CT

Before the introduction of PET/CT, essentially all multimodality clinical imaging was based on software fusion techniques⁷ and limited mainly to the brain. The introduction of the Hawkeye (GE Healthcare) in 1999, followed less than 2 years later by the first commercial PET/CT scanner, has irreversibly transformed the field of multimodality imaging. From 2001 to 2006, the sales of PET-only scanners decreased to 0, being replaced entirely by sales of PET/CT (Fig. 7). Currently, in 2008, a worldwide installed base of more than 2000 units attests to the rapid adoption of the modality by physicians.

The majority of this installed base is in routine clinical operation and there is, at least for oncology, a growing body of literature that supports the improved accuracy of staging and re-staging with PET/CT compared with either CT or PET acquired separately.^{102,103} Many of the pertinent publications have appeared within the past 2 to 3 years, and clearly document significant improvements in specificity and, to some extent, also in sensitivity, and especially in early detection of cancer recurrence.¹⁰⁴ These improvements are incremental when compared with PET, which alone demonstrates high sensitivity and specificity for a wide range of diseases. Improved accuracy has been documented for a variety of cancers including head and neck,^{58,105} thyroid,¹⁰⁶ lung,¹⁰⁷⁻¹⁰⁹ breast,^{110,111} esophageal,^{112,113} colorectal,^{114,115} and melanoma.¹¹⁶ There is also evidence that

PET/CT improves accuracy in lymphoma¹¹⁷ and solitary pulmonary nodules,^{118,119} despite the fact that in lymphoma the accuracy of PET alone is already very high.¹²⁰

In summary, therefore, the improvement in accuracy of PET/CT compared with PET or CT for staging and restaging is statistically significant and averages 10%-15% over all cancers.¹⁰³ As an illustration, Figure 8 shows 2 studies acquired on a Biograph 6 TruePoint TrueV PET/CT (Siemens Molecular Imaging) with a 21.8-cm axial FOV. Figure 8A shows transaxial PET and fused images of a 50-year-old woman diagnosed with pancreatic cancer. The images demonstrate intense focal uptake of FDG in a primary neoplasm 3.4 × 2.6 cm in size that can be accurately located in the head of the pancreas. No FDG uptake was identified in any of the proximal nodes although the likelihood of micrometastases would be high. Figure 8B shows a 58-year-old female patient with metastatic renal cell cancer. The images were acquired 110 minutes after the injection of 9.7 mCi of FDG. The total scan duration was 15 minutes with acquisition of 5 bed positions at 3 minutes per position. The CT was acquired at 130 kVp and 180 mAs (Siemens CAREdose). The study demonstrates a large FDG-avid peripherally enhancing necrotic mass occupying the anterior mid- and lower-left kidney. The mass is 10 cm in size and appears to involve the lower-pole collecting system.

Conclusion

There is little doubt that, during the past 6 years, PET/CT has had a growing impact on clinical imaging and particularly in oncology in staging and restaging disease and monitoring response to therapy. Although the technology has been somewhat disruptive in the sense that it has brought together medical specialties that have not traditionally worked together, namely, nuclear medicine and radiology, the overall

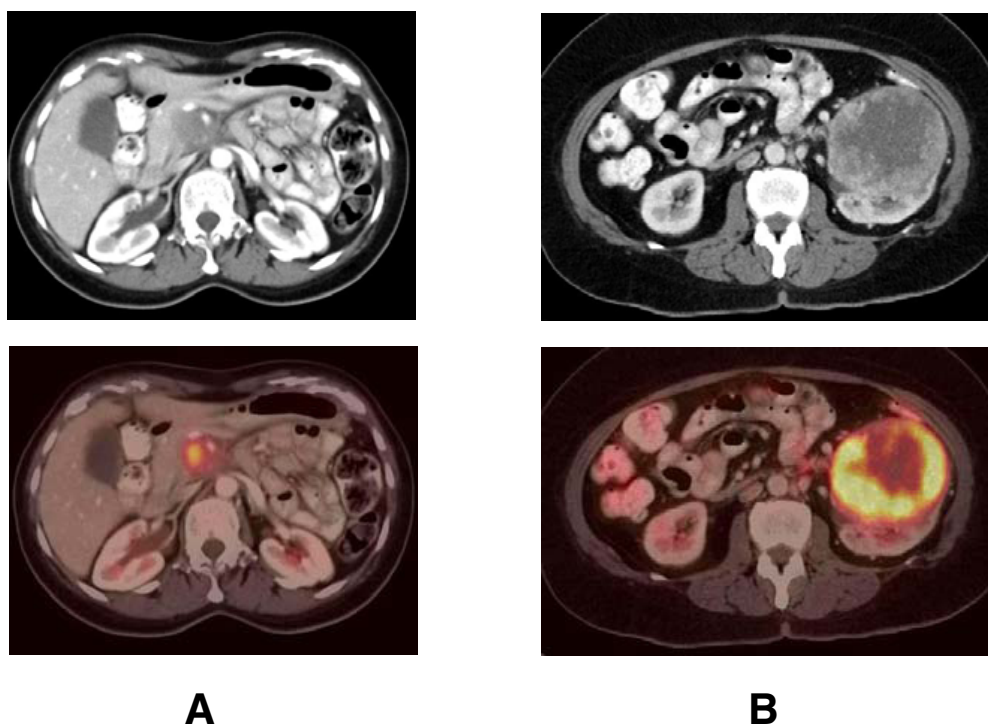


Figure 8 Two studies acquired on a Biograph 6 TruePoint TrueV PET/CT scanner. Transaxial sections are for PET (top row) and fused images (bottom row). (A) A 50-year-old woman with a diagnosis of pancreatic cancer. The images were acquired 94 minutes after injection of 10.3 mCi of FDG. The total scan duration was 10 minute with acquisition of 5 bed positions at 2 minutes per position. The CT was acquired at 130 kVp and 180 mAs (Siemens CAREDOSE). (B) A 58 year-old female patient with metastatic renal cell cancer from an unknown primary. The images were acquired 110 minutes after injection of 9.7 mCi of FDG. The total scan duration was 15 minutes with acquisition of 5 bed positions at 3 minutes per position. The CT was acquired at 130 kVp and 180 mAs (Siemens CAREDOSE).

impact has been positive. To meet the demand for cross-training of both the technologists who operate the devices and the physicians who interpret the studies, guidelines have been published¹²¹ and new standards established, leading to a somewhat different situation today from the way radiology and nuclear medicine have traditionally functioned. This trend is likely to continue as other multimodality devices reach the clinic, including SPECT/CT (introduced in 2004) and MR/PET (currently under development).

Acknowledgments

Dr. Townsend is a senior consultant with Siemens Molecular Imaging, Knoxville, TN, that manufactures and markets a combined PET/CT scanner.

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